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Privacy Risk Assessment of AI Agents in Open Environments

Félix Aguilar Sánchez

Supervisor: Patricia Callejo Pinardo

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Privacy Risk Assessment of AI Agents in Open Environments

Félix Aguilar Sánchez

Student, Carlos III University, Madrid, Spain,
100066993@alumnos.uc3m.es

Abstract—During recent years, Artificial Intelligence Agents have become the primary source for searching information on the Internet, replacing traditional web-search engines used in the last three decades. As their popularity grows, a new business model based on subscriptions and processing tokens has emerged, although 95% of the customers use a free-tier subscription. AI providers have benefited from this opportunity by targeting their users with personalized advertisements based on their preferences and the tailored profiles they developed. Nonetheless, this new scenario exposes a substantial gap between users’ privacy rights and the “privacy-by-design” frameworks under which these models claim to operate.

To address this issue, here we present a privacy risk assessment to audit the interactions of five real-world subjects across four popular AI agents. Our study evaluates the exposure of personal information from commercial profiling, comparing active memory context sessions with deactivated context memory scenarios, across five distinct metrics for all agents. The results reveal that, despite their alleged controls and safeguards, all evaluated agents leaked more than 20% of sensitive personal information, including: personal identifiers, ethnicity, orientation, religion, health status, or politics, among others. Furthermore, while half of the agents leaked, approximately 40% of the user’s data with the memory activated, this exposure escalated to more than 50% of the data in the short-term memory scenario. This counter-intuitive finding shows that deactivating session memory context does not prevent the generation of strong commercial profiling; rather, it amplifies data disclosure and worsens the privacy risk evaluation.

Index Terms—privacy risk assessment, data disclosure, GDPR compliance, AI agent, LLM-based AI agents

I. INTRODUCTION

Searching information on the internet via web-browsers has experienced a structural paradigm shift since the beginning of the millennium. Traditional web search engines such as Google have maintained the exclusive market dominance as the main access point to the Internet, traditionally relying on indexing keywords, user-driven link navigation, and search term frequencies. Today, autonomous Large Language Model (LLM)-based AI agents are replacing these ecosystems, in a way that interactions with computers have evolved into what has been defined as “Natural Language Programming (NLPg)” [1]. This progress represents a historical evolution in software abstraction: from traditional compilers to fixed-syntax languages, growing to Object-Oriented Programming (OOP), and ultimately consolidating into conversational natural language. As the software engineer Linus Torvalds recently observed, during all these years, software developers have never declared that all their code was written by compilers; yet, developers and companies consistently treat AI generation agents as an external entity that implements all their code, rather than the ultimate tool to exponentially improve their implementations and productivity [2].

As these AI agents become part of our personal, educational, and corporate environments, a new business model has emerged to maintain the increasing costs of data processing and resource consumption. Even though these platforms offer premium or pro subscriptions, approximately 95% of the user base relies entirely on the basic free-tier access [3]. To remain profitable, AI providers are indirectly turning these free layer subscriptions into profit by building highly granular and tailored customer profiles to target their users with directed advertisements. This new economic reality introduces critical data protection risks to user privacy that remains under review in actual cybersecurity literature.

This technological revolution completely redefines the digital privacy landscape of modern computer architectures. In the traditional web-model, user privacy exposure was limited to browsing history, web search queries or navigation cookies, profiling elements that required active tracking methods across every platform. In this new paradigm, agents do not really have a digital fingerprint per se; rather, users actively “gift” their complete profile information to an external provider. During the multiple conversations they maintain with commercial agents, whether for answering simple queries or solving complex problems, the shared information generates a critical disclosure of personal details through a unique, web-integrated interface. As a result, the leading exposure vectors of user’s identity have transformed from the digital local-device fingerprint to the agent’s active context memory, recent chat history and the information shared by the users.

Despite operating under a “Privacy-by-Design” (PbD) framework, commercial AI agent services present serious gaps between their announced safeguards and the operational reality. Our experiment unveils that the evaluated architectures follow a clear tendency to data disclosure, without compliance evaluations about privacy issues or data ownership, nor handling regulatory restrictions by any means. Furthermore, this weakness is not limited to user’s explicit information. The tangible threat lies beneath the surface, in the server-side infrastructure where AI-providers synthesize user’s unstructured information to infer protected attributes and sensitive data. Our thesis addresses these critical weaknesses by performing a systematic privacy risk assessment of two different context memory configurations, simulating the interaction between **five** real-world human subjects, through **twenty-two** conversational iterations with **four** state-of-the-art commercial AI agents: **ChatGPT** [4], **Copilot** [5], **Gemini** [6], and **Claude** [7]. Our comparison exposes the risks of structured customer profiling in short-term memory configurations and provides diverse metrics for compliant and privacy-preserving systems, trying to avoid more than the 40% of the personal data leaks shown in the obtained results.

II. RELATED WORK

The replacement of traditional Internet web-searches due to the increasing consumption of autonomous AI agents has generated a re-evaluation of current privacy and data protection frameworks. This section examines contemporary literature and the State of the Art across four different domains interconnected to our study: data privacy leakages of AI-driven ecosystems, the human-computer interaction towards AI-text generation, the behavioral and commercial profiling exploitation of AI-providers, and finally, the compliance and regulatory challenges imposed to these agentic infrastructures.

A. Privacy Leakages in AI-Driven Conversational Ecosystems

The assessment of data disclosures within AI-driven conversational ecosystems requires a systematic evaluation of multi-turn interactions to verify their existence. Despite recent monitoring studies proposed by Vekaria et al. [8], and browser extension audits focused on compliance protocols explored in AudAgent [9], the internal algorithmic profiling behavior of these models remains opaque to the final user. The Vekaria et al. [8] study establishes an important and critical baseline to analyze how third-party integrations and data-flows in LLM ecosystems are able to (un-)intentionally expose protected data and sensitive information. Complementing this network-view, AudAgent “introduces automated auditing methodologies to evaluate whether these agents satisfy the privacy policies under **PbD** frameworks, based on the leakages at a network-layer level” [9].

Information preservation constraints implemented across multi-conversational interactions introduce additional vectors for data exposure and sensitive information leakage. Modern studies have tracked different conversations with AI-chatbots [10], where conversational and memory context windows act as unstructured cumulative data repositories that allow AI providers to share personal information with third-parties, without users’ knowledge or explicit consent. This situation increases the model’s capacity to identity exposure, as the agents collect dispersed information across different conversations and compile it together into structured commercial user profiles. Other recent privacy-driven studies also showed important data leakage vulnerabilities in agentic architectures, inherent to long-term memory configurations [10].

Finally, the published *LeakyLM* auditing framework has demonstrated that “Your AI assistant is leaking your conversations” [11]. Commercial agents capture user’s conversational interactions and unstructured descriptions to later disclose personal information and chat metadata to external tracking domains. Unlike traditional tracking methods that relied on static cookies or advertising analytics, user’s explicit selections on cookies consent have no real effect on the information these agents share with third-party trackers. [11]

B. Human-Computer Interaction and AI-Text Generation

Recent publications in computational linguistics reveal that user interactions with generative AI models substantially alter the nature of data generation and exposure [12]. A structural paradox is also created when the automatic text generation dilutes authorship and data intellectual property boundaries and transforms this automatization process into a “generation of large amounts of low-quality data that nobody writes and nobody will end up reading” [12].

This paradigm shift triggers that users no longer control the data workflow lifecycle of the messages they generate, since the data is passively collected during the conversational interactions. The increasing integration of these generative models within our daily workflows transforms user conversations and data inputs into training data and profile elaboration for these models, altering the traditional information boundaries, data security, privacy and even the meaning of the data itself [12].

The evolution on human-computer interaction requires a deeper analysis of the algorithmic and technical nature of these generative models, from an ethical and cybersecurity point of view. Academic research on the integration of Artificial Intelligence within personal and educational environments highlights that LLM models are incorporated into the user’s daily lives and no longer limited to simply recover user data. Instead, they synthesize and process behavioral patterns based on their conversational interactions with the users, their personal choices, and topic preferences [13]. Consequently, privacy risks are no longer limited to explicit data inputs, they are expanded to the generation of automated responses. This situation leads into “user profiling and uncontrolled data disclosures when the model adapts to the historical context”, as noted by legal and technical scholars [14].

Privacy preservation across multi-turn iterations introduces additional data exposure vectors, influencing the user’s willingness to disclose personal and sensitive information. As explored by Kwesi et al. [15], AI agents “are increasingly being appropriated for emotional support and companionship” which determines how the users interact with conversational agents and generates new challenges for security and privacy controls, such as model training data exclusions or granular preferences on memory configurations. Their findings reveal that the evaluated memory controls of the agents do not completely delete the algorithmic inferences within the system, they affect the persistent context-memory configurations and the privacy and security boundaries, determining real-world data disclosure risks [15].

C. Behavioral and Commercial Profiling Exploitation

User privacy and data protection have always been confronted with the economic interests of technological corporations, and AI-providers are no exception to this reality. Recent studies on user profiling assess how free-tier subscription models generate business incentives allowing these companies to commercialize conversational telemetry, transforming this monetization into a corporate strategic priority. The challenge to address now is users’ privacy “and trustworthy intervention in the generative process, not about content placement”. [16]

Because free-tier subscriptions completely overcome paid-subscriptions [3], AI providers optimize data consumption to extract user preferences and relevant information for targeted advertising purposes. However, this monetization strategy generates an unexpected drawback as a consequence: natural language processing becomes an exposure vector because advanced statistical correlations are used to infer implicit user characteristics, and these relationships are used to generate tailored and highly accurate commercial profiles [16]. This situation is transforming the user conversational environment into a dedicated marketing channel for advertisers to reach higher potential customers and improved targeted audiences [16], [17].

In fact, major chatbot providers such as **OpenAI** [18] or **Microsoft** [19] have recently inserted commercial ads within their agent’s conversations, while Google **Gemini** team is currently working on “help brands connect with consumers” [20]. This specific risk regarding commercial profiling has been corroborated by academic research, including studies from the University Carlos III of Madrid, where different methodologies demonstrated that commercial profiling in web environments has been successfully migrated into commercial LLM-based agent frameworks. “Behavioral advertising, driven by AI algorithms, manipulates subconscious decisions, invades privacy, and increases data security risks, leading to digital fatigue and mental health problems such as anxiety and compulsive consumerism” [21]. The analysis of the semantic composition of agent conversations across multi-turn interactions has confirmed that automated profiling systems are able to accurately map user preferences, socioeconomic statuses and consumer habits.

These predictive and generative capacities cause traditional **PbD** frameworks to remain ineffective against privacy and data disclosure, as the risk merges from a simple retention policy to a complex profiling heuristic that challenges user’s fundamental rights to data protection, privacy and control. Related studies argue that “generative AI fundamentally changes advertising: rather than placing products into discrete slots, it enables interventions on the generative process itself, which induce commercial influence through less observable channels” [16], transforming generative AI advertising into a challenging new paradigm.

D. Privacy Regulation and Data Minimization Compliance

The process of commercial predictive profiling often fails when it is audited under a regulatory framework. According to the European **GDPR** (“*General Data Protection Regulation*”) [22] and the Spanish transposition **LOPDGDD** (“*Ley Orgánica de Protección de Datos Personales y Garantía de los Derechos Digitales*”) [23], “data processing operations must strictly comply with both purpose limitation and data minimization principles”, which conversational AI architectures typically neglect [8], [10], [11]. When the agents maintain a multi-session persistent memory configuration, or accurately infers missing information to fulfill a structured *JavaScript Object Notation* (JSON) template (as the one provided in our assessment), these models frequently ignore the data collection and minimization thresholds, actively bypassing all these principles and regulations [24].

Handling special categories of personal and sensitive information creates an additional vulnerability layer. Legacy configurations implemented rigid database schemas and input validations to filter sensitive and protected attributes, but LLM-based agents process this information dynamically within their conversational context. The European Union’s **AI Act** [25] imposes rigorous obligations on systems that process sensitive information or perform automated commercial profiling, which current commercial deployments do not implement. This lack of explicit verification mechanisms prior to the personal attribute inference creates a measurable gap between corporate privacy declarations and their effective implementations, as we will illustrate in subsequent sections.

III. METHODOLOGY

To evaluate the privacy implications and profiling capabilities of state-of-the-art conversational ecosystems, our study implements an experimental auditing framework, aiming to assess the interaction boundaries between human users and conversational Artificial Intelligence agents. All these interactions are represented by **five** real-world subjects, interacting with **four** distinct commercial LLM-based AI platforms, over **twenty-two** sequential prompt iterations for each individual subject-agent pair. This methodology was designed to establish a robust and complete trace of the conversational telemetry during all the conversations. This design generates a consolidated dataset of processed natural language, examined methodically through five exposure metrics: Direct Recovery (**DR**), Plausible Inference (**PLI**), Hallucination (**HA**), Data Leak (**DL**) and Sensitive Field (**SF**). These evaluation parameters will allow us to classify how autonomous agents retain, process, and disclose implicit or explicit user data during multi-turn interactions, and they serve as leading indicators of the analytical phase and results of this study.

Five individual profiles were created to perform this experiment, in order to simulate a real human-agent conversational interaction. This subject group represents a multi-generational distribution of people, inspired by the primary researcher (**Subject 1**) and his family (**Subjects 2 to 5**). The parameters used to define all experimental subjects were fabricated without incorporating real-world or identifiable individual data. No additional data protection controls were needed because all profiles were completely simulated and lacked both personally identifiable information (PII) and direct specific identifiers. Accordingly, our experiment consistently satisfies the cybersecurity, privacy and ethical standards established by the GDPR [22], [23] and AI Act [25] regulations, successfully complying with its **PbD** [26] and personal data processing [27] principles. The verification of the obtained results was performed through a feedback evaluation process, where the primary researcher thoroughly reviewed and validated the accuracy of the profiles generated by the agents against each subject conversational input, ensuring that the structural integrity of the experiment was consistent and represented all five subjects correctly.

The following subsections explain the architectural environment; agent and browser configurations; datasets and subject profiles, and the metrics utilized to quantify the data exposure, proving the obtained results of the assessment.

A. Agent Selection and Infrastructure

As previously stated, the scope of the experiment targets four of the most widely deployed commercial AI Agent services: OpenAI’s **ChatGPT** [4], Microsoft’s **Copilot** [5], Google’s **Gemini** [6], and Anthropic’s **Claude** [7]. These platforms represent the most used consumer artificial intelligence agents, each operating and maintaining the user’s configuration settings for each of the agent’s context and memory retention [28], [29], [30], [31].

To simulate an authentic interaction environment and capture as realistic telemetry data as possible, we created five independent user profiles (one for each Subject). Each Subject was registered to a dedicated Google account, with a unique Gmail address, and using a real-world cellular phone verification process for the platform registration and authentication.

After the account registration, each user accessed each platform by linking and authorizing their Gmail account. This configuration simplifies the signing up process and perfectly simulates the actual behavior of a regular user. The Gmail account authorization is also useful for our assessment, because with this connection we can analyze which data the agents are obtaining from each email account. The interaction across agents was performed through different user profiles and different web browsers to avoid cookie or cache contamination between subjects and browsers. This environment isolation perfectly aligns with the established criteria in other contemporary web-assistant audits [8]. All conversations and data defined in our study were chosen to simulate regular iterations with an agent: help with daily tasks, recommendations on their work life, or different advice about personal issues.

Additionally, the local setup included specific configurations to avoid inaccurate information during the evaluation and commercial profiling. To simulate the real residence of each subject, a local Virtual Private Network (VPN) was used on the device (and the web browser) of each subject, with the location established on the municipality defined for each subject (or a closer one). In this sense, and from the agent point of view, **Subject 2** and **Subject 3** were connected from their simulated cities, and the other subjects were connected near their locations (South of Madrid, Toledo and surroundings). Simulating a native residential network connection ensured that local-side browser trackers and the service providers of the agents captured authentic geographic coordinates and local routing metadata. The selected configuration allowed us to assess, in the most objective manner, how all shared metadata through the network interacts with the internal memory and context of the agents, providing a realistic evaluation of the data minimization and **PbD** frameworks these agents supposedly implement and comply with, as studied in contemporary privacy auditing literature [8], [9], [10].

B. Operational Models, Memory and Browser Configuration

We established an experimental framework based on a two-scenario testing protocol, to evaluate how the agents process and profile the shared conversational data under different configurations. This design is divided into two operational states, executed across all four agents: the mode with the memory activated (or **Active Memory State**), and the mode with the memory deactivated (or **Inactive Memory State**):

- Under the **Active Memory State (AMS)**, each platform context retention and memory references from other conversations were activated (Fig. 1 and Fig. 2). This configuration simulates a standard setup where the agent dynamically updates and consolidates its internal context across independent conversations, in a regular session.
- The **Inactive Memory State (IMS)**, on the contrary, establishes a no context baseline for each agent. In this mode, context and memory persistence features were explicitly deactivated to isolate the agent’s conversation-related inference capabilities, allowing us to quantify whether commercial preference harvesting and profiling persist, despite context and memory minimization. As previously stated, this configuration was activated after the **AMS** assessment was completed.

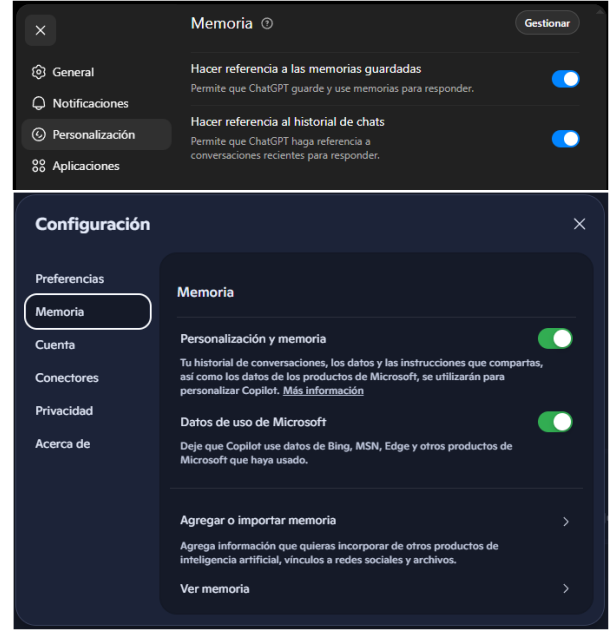


Fig. 1. ChatGPT and Copilot Active Memory Configuration.

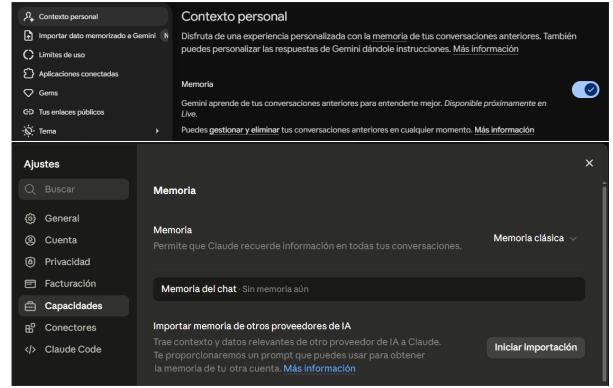


Fig. 2. Gemini and Claude Active Memory Configuration.

For all five experimental subjects, interactions across each agent service was restricted exclusively to the default, basic tier model variants. At the time this study was written, the selected free available model for each agent is shown in the following table:

ChatGPT	Copilot	Gemini	Claude
<i>GPT-4o mini</i>	<i>Smart</i>	<i>3.5 Flash</i>	<i>Sonnet 4.6 Low (Non-reasoning)</i>

Table I. Agent’s Selected Models

This subscription-free constraint deliberately excludes premium processing parameters and high-throughput context window upgrades, while evaluating the compliance of each agent with the **PbD** framework, from the experience of a non-subscribed consumer. Standardizing the execution tier to basic/free models ensures that the tracking behaviors, data disclosure metrics, and profiling vectors captured during the **twenty-two** iterations represent the genuine privacy boundaries encountered by everyday users who interact with these conversational systems, without specialized administrative privileges or monthly subscriptions.

Finally, to prevent system-side bias deriving from any browser architecture’s privacy policies, or cookie-handling frameworks, each individual subject profile was bound to an independent web browser environment. This distribution allows our evaluation to analyze how peripheral client-side systems interact with each conversational agent services.

Subject 1 completed all conversational interactions using the Brave browser, an environment that enforces strict, native tracker and script blocking by default. **Subject 2** was configured to execute all interactions via Microsoft Edge, representing standard corporate configurations operating on a Chromium core architecture. **Subject 3** performed all cycles within Mozilla Firefox, representing a user which prioritizes privacy and open-source developments. **Subject 4** utilized the Opera web browser, employing its client-side security tools. Finally, **Subject 5** was evaluated utilizing Vivaldi, a Chromium-based platform optimized for advanced user control over cross-site cookie isolation and storage retention rules. This heterogeneous configuration ensures that the quantification of data leaks and profiling accuracy captures authentic platform-side telemetry across a representative cross-section of modern consumer web infrastructure.

Subject	Browser
Subject 1	<i>Brave v1.92</i>
Subject 2	<i>Microsoft Edge v150.0.4</i>
Subject 3	<i>Mozilla Firefox v151.0.5</i>
Subject 4	<i>Opera v133.0.59</i>
Subject 5	<i>Vivaldi v8.1.40</i>

Table II. Subject’s Web Browsers

C. Data Taxonomy and Conversational Protocols

We structured all the subject’s inputs across a structured framework to evaluate the generation of each profile effectively. Prior to the data ingestion, all attributes were organized into seven independent operational categories: Personal, Demographic, Lifestyle, Socioeconomic, Physical & Wellness, Health/Sensitive and Technological. Each category initially comprised a baseline matrix ranging from 7 to 17 potential attributes designed to capture a comprehensive representation of an individual user’s lifestyle; even though, the final assessment did not require the complete set of attributes and categories. Table III illustrates these categories and examples.

From this taxonomical baseline, the information was divided into implicit telemetry data and explicit conversational disclosures provided by the users. The implicit data represent structural metadata natively accessible to the agent during an active session, meaning: the user’s language, account identifiers, custom avatar, vocabulary, writing style, and session-specific timestamps. This category also captures low-level network infrastructure details and browsing metadata, including network location, IP address, GPS approximate location, operating system, browser name and device fingerprinting variables. On the other hand, explicit data information included a standardized profile template of 26 attributes deployed across each subject interaction. This template included several indicators such as the ones previously shown on Table III as an example, among many others.

To establish a rigorous validation control layer within the auditing architecture, two critical attributes were intentionally

undisclosed and entirely excluded from the conversational interactions with all the agents: each subject’s **marital status** and the **date of birth** (only available during the initial account registration). No explicit statements, relational summaries, or implicit linguistic clues regarding the user’s civil state were introduced at any stage of the conversational cycles, providing a clean baseline to evaluate the heuristic inference capacity of the models.

The prompting execution phase inserted these attributes in the conversational interactions. For **Subject 1**, **Subject 2**, **Subject 3**, and **Subject 4**, the interactions were separated into **22** distinct prompt iterations across **9** independent, thematic conversational threads to ensure clear context tracking. These thematic blocks simulated real conversational scenarios of any standard user, including:

- 1) CV optimization for professional improvement.
- 2) Home and schooling location guidance.
- 3) Family hobbies and recommendations.
- 4) Health status suggestions.
- 5) Financial advice on buying vs. renting a car.
- 6) Irrelevant questions about unrelated topics (the capital of a country, a random number, a pasta recipe or a simple explanation about a different subject).
- 7) Simple inference and recalling tests based on the agent’s context and memory.
- 8) Generation of the subject’s commercial profile.
- 9) Privacy and legal compliance self-assessment.

During the last two iterations (conversations 8 and 9), a JSON file template was provided to each agent for the complete evaluation and final quantification of the results. The file template is shown in Appendix B of this study.

To compare the memory configuration and commercial profiling process with other subjects, **Subject 5** introduces a slight variation in the conversational flow. This variation was thoroughly performed to assess if a unique conversation in contrast to iterating on separated conversations alters the final results of the commercial profiling. In this scenario, with this subject, the dialogue was performed under a unique conversational flow, without referencing other chats. In addition to this, the web-browser used in this iteration was configured to implement zero-tracking policies by default. This setup was used to actively reject non-essential tracking cookies and block commercial scripts, aiming to isolate the influence of peripheral browser telemetry from the text processing and conversation with the agent.

After all conversational iterations on the **AMS** configuration were finished, the auditing protocol transitioned into a reset memory phase. For three days, no subject interacted with any agent. This interval was established to ensure that the agent’s context memory cooled down completely and previous iterations left no trace in the system. After this time interval, a fresh new session was initialized upon each subject with each agent, in a separate and different conversational iteration. To maintain the consistency with the previous AMS phase, the same nine-conversational workflow sequence was followed in identical order, and the final extraction of the structured commercial user profile (according to our predefined JSON template), was performed in last iterations 8 and 9.

Category	Sample Values							
Personal	name	email	phone	username	avatar	birthdate	age	civil state
Demographic	city	zip code	country	state	location	town	street	area
Lifestyle	hobbies	travel	food	movies	TV shows	videogames	books	brands
Socioeconomic	education	vocabulary	income	profession	sector	position	company	children
Physical & Wellness	weight	height	hair	constitution	activity	disability	body fat	BMI
Health/Sensitive	health	religion	politics	union	race	ethnicity	orientation	affiliations
Technological	IP	location	coordinates	OS	device	fingerprint	browser	timestamp

Table III. Categories of Information Collected and Attributes Selected During the Experiment

D. Profiling Quantification and Regulatory Self-Assessment

Using several industry standards on marketing profiling as guidance [32], [33], [34], and two online resources available as example [35], [36], we designed a standardized JSON data extraction template to facilitate a structured and objective quantification of the complete conversational telemetry for each agent and subject. This template represents a complete user profile, structured into **fourteen** different categories under which all user fields and attributes will be evaluated:

- 1) Baseline identity metrics (*identity*)
- 2) Inferred demographic data (*demographics_inferred*)
- 3) Client's device profile (*device_profile*)
- 4) Professional activities (*professional_inference*)
- 5) Educational milestones (*education_profile*)
- 6) Financial parameters (*financial_profile*)
- 7) Family context (*family_context*)
- 8) Commercial segments (*commercial_segments*)
- 9) Category-specific interests (*interest_segments*)
- 10) Consumption preferences (*consumer_profile*)
- 11) Holistic lifestyle summaries (*lifestyle_summary*)
- 12) Product relevance thresholds (*product_interests*)
- 13) Behavioral modeling markers (*behavioral_model*)
- 14) Interest probabilities (*interest_probabilities*)

Crucially, the schema embeds an important control meta-data object defined by **18** compliance attributes within the *extra_notes* category. This node enforces an explicit definition of the overall profiling and a qualitative description of its limitations. In this block, the audited agents are commanded to perform a complete data quality self-evaluation, assigning an internal confidence score to all attributes derived from its algorithmic inference. At the same time, each agent was required to perform an assessment of its own limitations and confidence over their inferences, arguing about the quality of each subject's obtained data, and assigning a percentage score of each data category for the quantitative measurement.

Beyond consolidating the profile of each subject, the extraction protocol generates a dedicated privacy evaluation section to audit the alignment of each system with Spanish and European data-protection regulations [22], [23]. Agents autonomously evaluate the compliance of their data-ingestion pipelines by populating the regulatory-related fields *rgpd_compliance* and *compliance_score*, for its related score and rationale. This assessment evaluates agentic execution parameters against the core principles of data minimization, purpose limitation, and lawful-basis processing as mandated by GDPR and LOPDGDG regulations, as discussed in Subsection II-D.

From an operational point of view, obtaining these fields required a high degree of prompt persistence and dialogue insistence during the final interactive window. The architectural and risk implications of this behavioral divergence are examined in depth in the subsequent results chapter.

IV. RESULTS

This section presents a quantitative and systematic analysis of the results obtained during our empirical assessment. The resulting matrix evaluates **142** attributes, across **5** human subjects and **4** autonomous agents. The output dataset was generated after processing and extracting the complete **22** subject-agent conversational multi-turn interactions. As a result of all these combinations, **2,840** attributes were evaluated in total (excluding last **18** *extra_notes* compliance metadata).

A. Taxonomical Distribution and Profiling Metrics

The complete telemetry obtained from each subject's profile was classified according to the five exposure metrics explained on Section III: **D**irect **R**ecovery, **P**lausible **I**nference, **H**allucination, **D**ata **L**eak, and **S**ensitive **F**ield metrics. The empty attributes within each classification were represented as **No Data (ND)**.

JSON Category	ChatGPT	Copilot	Gemini	Claude
<i>identity</i>	DR	DL	DL	PLI
<i>demographics_inferred</i>	PLI	PLI	HA	DL
<i>device_profile</i>	DL	DL	PLI	HA
<i>professional_inference</i>	DR	DR	PLI	PLI
<i>education_profile</i>	HA	PLI	DR	DR
<i>financial_profile</i>	DR	HA	PLI	DR
<i>family_context</i>	DR	PLI	DL	HA
<i>commercial_segments</i>	PLI	DR	HA	PLI
<i>interest_segments</i>	HA	DR	PLI	DR
<i>consumer_profile</i>	DR	HA	DR	PLI
<i>lifestyle_summary</i>	DR	PLI	HA	PLI
<i>product_interests</i>	PLI	HA	DR	DR
<i>behavioral_model</i>	HA	DR	PLI	HA
<i>interest_probabilities</i>	PLI	PLI	HA	DR

Table IV. Profiling Metrics by Schema Category

Table IV shows a summarized example of the data quantification for each JSON template category, although it is essential to emphasize that the evaluation targeted each specific attribute within each category, rather than the category per se.

To determine the tracking capabilities of every model, and how context-memory setup impacts the user profile exploitation, all fourteen categories were analyzed under both the **AMS** and the **IMS** configurations. After an initial assessment, the resulting dataset indicated that these commercial LLM-architectures prioritize semantic synthesis and attribute composition over data minimization. Across all subjects, every agent produced tailored representations for commercial profiling purposes, and they systematically and consistently aggregated unstructured conversational data into structured behavioral profiles, regardless of their information or environment configuration,

The analysis of the complete dataset highlights a severe structural trade-off between contextual responsiveness and data minimization principles. When operating under the default **AMS** configuration, all audited agents actively preserved historical identifiers and combine implicit telemetry signals with explicit natural language disclosures. The global results for the complete population of this configuration are presented in Table V. Each cell of the table shows the cumulative count of the audited attributes, resolved across all five profiles for each agent, arranged by exposure metric:

Exposure Metric	ChatGPT	Copilot	Gemini	Claude	Total
Direct Recovery	194	159	170	200	723
Plausible Inference	277	258	262	318	1115
Hallucination	193	236	216	152	797
Data Leaks	31	39	44	21	135
No Data	15	18	18	19	70
Total Attributes	710	710	710	710	2840

Table V. Profiling Exposure Absolute Metric under AMS

The baseline distribution shows that explicit attribute retention via **DR** metrics is heavily dependent on the specific interaction with the agent. **Claude** demonstrated the most aggressive indexing architecture, successfully extracting **200** direct disclosures out of the **710** total population attributes (28,17%). This performance was followed by **ChatGPT** with **194** direct resolutions (27,32%), while **Gemini** and **Copilot** recovered **170** and **159** direct attributes respectively (under 24%). This variance indicates that while some platforms rely on immediate string indexing, others employ aggressive parsing layers to map context variables.

This tracking capacity becomes critical when we analyze the inferred data of each profile through **PLI** metric. Across the entire demographic domain, accurate deductions regarding unshared consumer preferences and lifestyle constraints represented the largest category of resolved profile variables (a distribution of more than 36% of the attributes for every agent). **PLI** resolutions peaked significantly in **Claude** with **318** successful inferences (44,79%), representing the higher profiling success rate based purely on inferred correlations. In comparison, its competitors generated **277** (**ChatGPT**, 39,01%), **262** (**Gemini**, 36,90%), and **258** (**Copilot**, 36,34%) plausible inferences across the global dataset attributes, as shown on Table V and Fig. 3. This pattern confirms that the absence of explicit identifiers does not inhibit profile generation; rather, the agents successfully bypass user omission layers by cross-referencing implicit contextual clues scattered across unrelated conversational iterations.

However, these commercial and predictive optimization responses introduced a significant and structural data degradation through the increase of generative hallucinations (**HA**). In the presence of missing variables within a predefined schema, models generate plausible but false information to complete inferred profiles. This generation of inaccurate and fabricated attributes was clearly prominent in **Copilot** and **Gemini**, which registered **236** (33,24%) and **216** (30,42%) hallucinations respectively. In contrast, **Claude** presented the lowest hallucination rate (21,41%), limiting the invented fields to **152** attributes and indicating a more conservative threshold when performing heuristic profile generation. **ChatGPT** established the balance among the agents with **193** hallucinations in total (the 27,18%).

Finally, the cumulative security risk is highlighted by tracking context data disclosure via **DL** metrics. These information leaks represent samples where conversational metadata or sensitive information crossed session perimeters and became accessible to the backend corporate server. The cumulative data leaks across the entire subject group reached a maximum of **44** instances for **Gemini** (6,2%) and **39** for **Copilot** (5,49%), while **ChatGPT** and **Claude** registered **31** and **21** leak vectors, (4,37% and 2,96%) respectively. The persistent appearance of high **DL** indicators across all five subjects (**135** attributes among all agents, almost a 5% rate) validates that consumer-facing conversational frameworks operate as proactive telemetry-harvesting systems.

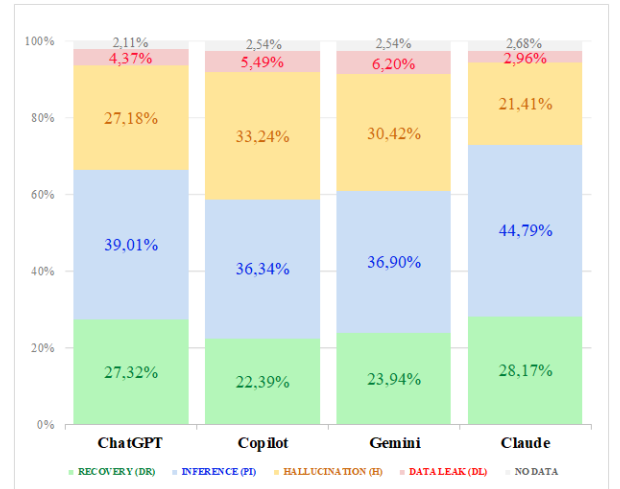


Fig. 3. Profiling Exposure Distribution Metrics under AMS

In practice, these architectures transform regular and unstructured conversational iterations into commercial profiling data, creating an essential non-compliance conflict against the main principles of purpose limitation and **PbD** frameworks. As previously discussed, this situation is a great commercial opportunity to AI providers for consumer profile trading with external third-parties for advertising purposes [8], [10], [11], [16].

Fig. 3 presents the distribution metrics of these indicators per agent, revealing that all models inferred and hallucinated **over 50%** of the attributes provided in the evaluation schema. Unfortunately, these indicators will worsen under the **IMS** configuration, as we will show in following Subsection IV-B.

B. Context Retention and Memory Configuration Impact

This subsection presents a detailed assessment of the context window retention and memory configuration impact across all agents. Specifically, we contrasted the **AMS** vs. the **IMS** scenario to evaluate how memory deactivation drives performance degradation for each individual subject. After a quantitative comparison between the two operational configurations, the results revealed a drastic degradation and a paradigm shift in heuristic profiling metrics when memory deactivation is enforced.

As performed during the previous **AMS** evaluation, we analyzed the systemic impact across **DR**, **PLI**, **HA** and **DL** dimensions, under the **IMS** configuration across all agents, based on the data gathered from the same sample of **710** attributes per model, as illustrated on Table VI:

Exposure Metric	ChatGPT	Copilot	Gemini	Claude	Total
Direct Recovery	89	25	23	50	187
Plausible Inference	532	481	486	540	2039
Hallucination	7	14	24	9	54
Data Leaks	25	53	55	33	166
No Data	57	137	122	78	394
Total Attributes	710	710	710	710	2840

Table VI. Profiling Exposure Absolute Metric under IMS

The first critical finding when transitioning to **IMS** is the widespread collapse in the agents’ ability to perform a **DR** metric of exacted attributes, while almost duplicating the number of inferences. Evidence shows a clear degradation on the **DR** metric: upon this configuration the total **DR** attributes drop drastically from **723** to **187**, a **74.14%** reduction. **Copilot** and **Gemini** exhibit the highest vulnerability to memory loss, reducing their exact recovery capacity to a marginal **3.52%** and **3.24%** respectively.

The **DR** reduction produced an inference compensation effect over the **PLI** metric because, in the absence of accessible persistent memory logs in **IMS**, agents mitigate the information vacuum by spiking their **Plausible Inference** metrics. The total volume of inferences rises from **1115** in **AMS** to **2039** occurrences in **IMS**, nearly **twice** the occurrences in this configuration. **ChatGPT** nearly doubles its inference rate, jumping from **39.01%** to **74.93%**. **Copilot** and **Gemini** follow an identical trend, reaching **67.75%** and **68.45%**, respectively. In the case of **Claude**, although the relative record indexes at **76.06%** due to heuristic classification deviations in the response rate, its absolute volume scales forcefully up to **540** instances. These results reveal that commercial agents do not explicitly acknowledge missing user information. Without setting up a persistent context memory configuration, these agents transition to a simulation and predictive generation mode, which substantially alters profiling reliability.

One of the most concerning aspects from the profile integrity perspective is the **HA** behavior in relation to the memory state. The results of the assessment illustrate an operational paradox: while intuitive assumptions suggest that an **AMS** configuration will provide stability to the agent, the quantitative results show that this same state serves as the primary catalyst for a higher number of hallucinations.

While the total volume of **HA** in **AMS** reached **797** cases, only **54** documented cases emerged in the **IMS** configuration, representing a **93%** reduction. Driven by the increase in plausible inference rates and simulated data attributes, the **HA** rate fell to **0.99%** for **ChatGPT**, marking the lowest value among all agents. In contrast, **Gemini** exhibited the highest **HA** index under this configuration at **3.38%**, followed by **Copilot** at **1.97%**. Under these same conditions, even **Claude**, typically the most conservative model, displayed a marginal **HA** rate of **1.27%**.

The comparison of the dataset results between both memory configurations revealed that **AMS** hallucinations are not purely semantic attribute variations. An evident trend towards the notorious inventions of highly specific biographical data is observed. The agents erroneously infer and assert with absolute certainty information that were not present during any agent-subject interaction (e.g., inventing explicit data regarding alleged children or family status of the subjects) and without any relation with the conversation history. This state of “*Hallucinatory Delirium*” suggests that context window retention mechanisms fail and override the defined safeguards on memory boundaries during long-time conversations. The situation generates fictitious user profiles that are erroneously structured as facts and severely downgrades the data quality of the agent’s knowledge base. Under **IMS** configurations this trend disappears, since a smaller volume of prior knowledge minimizes the opportunities for hallucinations, although it duplicates the inference rates.

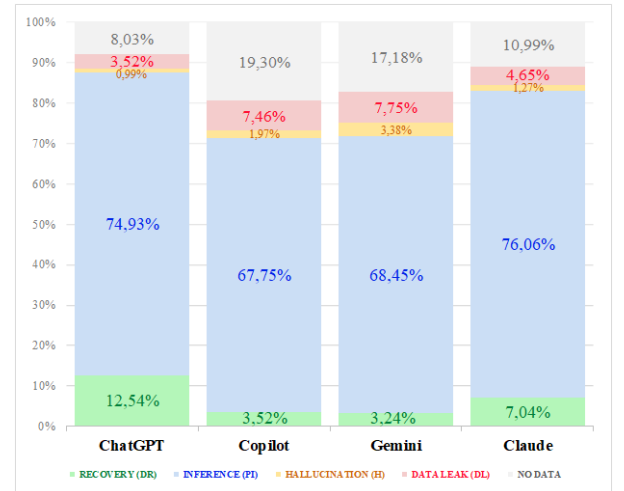


Fig. 4. Profiling Exposure Distribution Metrics under IMS

From the perspective of data privacy and cybersecurity, the **DL** metric dimension provides the most concerning conclusions of this comparative study. There exists a latent risk in **IMS** configurations because it is traditionally assumed that disabling memory should protect user’s privacy. However, empirical data contradicts this hypothesis for most evaluated models. Upon disabling memory, data leaks significantly increase up to **75%** of the audited agents. **Copilot** increases its leak rate from **5.49%** in **AMS** to **7.46%** in **IMS**. **Gemini** suffers a similar degradation, raising its leaks from **6.20%** to **7.75%**. **Claude**, on its part, increases its vulnerability from **2.96%** to **4.65%**. Only **ChatGPT** shows slight containment in this comparison, reducing its leaks from **4.37%** to **3.52%**.

These metrics suggest that, when a consolidated memory state is not configured properly to act as a persistent safeguard filter, an agent with an **IMS** can be more susceptible to social engineering, role-play or prompt injection attacks that extract sensitive data from the conversation. Eventually, the **No Data (ND)** metric confirms the practical degradation of the **IMS**: while the absence of data represented only a **2%** rate in the **AMS**, the total volume of empty or null responses rises from **70** in **AMS** to **394** in **IMS**. Microsoft’s **Copilot (19.30%)** and Google’s **Gemini (17.18%)** blocked the output rather than risk a response in nearly a fifth of interactions when lacking active memory. Anthropic’s **Claude** and OpenAI’s **ChatGPT** reduce these rates to **10.99%** and **8.03%**, respectively.

Profiling Degradation Index (PDI)

We define the *Profiling Degradation Index* (PDI_m) to mathematically quantify the degradation factor experienced by any exposure metric (m). This calculation is represented in relation with the **DR** variation, and during the transition from the **AMS** to the **IMS** configuration, as follows:

$$PDI_m = \frac{|\Delta MD|}{|\Delta DR|} \in \mathbb{Q} \quad (1)$$

where the MD value represents the evaluated **Metric Dimension**, and the delta operators (Δ) represent the structural variations between both memory configurations:

- $\Delta \text{PLI} = \text{PLI}_{\text{IMS}} - \text{PLI}_{\text{AMS}}$: variation in **PLI** occurrences.
- $\Delta \text{HA} = \text{HA}_{\text{IMS}} - \text{HA}_{\text{AMS}}$: variation in **HA** occurrences.
- $\Delta \text{DR} = \text{DR}_{\text{IMS}} - \text{DR}_{\text{AMS}}$: variation in **DR** capacity.

The calculation will be performed over **PLI** and **HA** dimensions, measuring two different indexes. **DL** and **ND** dimensions are intentionally excluded because they represent privacy behaviors analyzed in detail in Subsection IV-C.

By substituting the total empirical measurements obtained throughout this study ($\text{PLI}_{\text{IMS}} = 2039$, $\text{PLI}_{\text{AMS}} = 1115$, $\text{HA}_{\text{IMS}} = 54$, $\text{HA}_{\text{AMS}} = 797$, $\text{DR}_{\text{IMS}} = 187$, and $\text{DR}_{\text{AMS}} = 723$), the proposed model results:

$$PDI_{\text{PLI}} = \frac{2039 - 1115}{187 - 723} = \frac{924}{536} \approx 1.724 \quad (2)$$

$$PDI_{\text{HA}} = \frac{54 - 797}{187 - 723} = \frac{-743}{536} \approx -1.386 \quad (3)$$

Using the **AMS** configuration as the reference baseline, the obtained values $PDI_{\text{PLI}} = 1.724$ and $PDI_{\text{HA}} = -1.386$ quantify the behavioral degradation, produced in each dimension after disabling the persistent memory. For the **PLI** dimension in particular, this value represents a **1.724** increment factor of the plausible inferences respect to the **DR** dimension. On the other hand, the **HA** experienced a negative **-1.386** factor degradation in hallucinations respect to **DR**.

PDI_{PLI} and PDI_{HA} aim to quantify the behavioral factor degradation produced after disabling persistent memory. Rather than measuring the absolute (or percentual) reduction of **DR**, the PDI expresses the variation factor of each different dimension **PLI** and **HA**, related with the variation of the **DR** dimension. These calculations support distribution transition between **AMS** and **IMS** configurations, as illustrated on Fig. 3 and Fig. 4 respectively.

From the data privacy and cybersecurity perspectives, these results refute the assumption that simply disabling persistent memory improves user’s privacy. Although the **IMS** configuration substantially reduces hallucinations (in a **-1.386** factor related to **DR** variation), it severely weakens the profiling reliability and increases the number of plausible inferences (in a **1.724** factor related to **DR** variation). The **IMS** configuration increases agentic inferences about user behavior and commercial profiling, whereas it worsens accidental data exposures and the application of contextual memory safeguards.

C. Client-Side Hardening and Data Disclosure

This section evaluates the operational limits of client-side defense mechanisms and their real impact on the privacy boundaries of commercial AI agents. Specifically, we examine whether enforcing client-side browser hardening configurations (blocking commercial tracking, scripts, and third-party cookies), affects LLM-agents to collect profiling information and disclose personal data. To provide a client-side security assessment, we examined the disclosed sensitive variables under a scenario with the browser-level controls activated, against the same results with the browser-level controls deactivated.

1) *Client Safeguard Paradox*: A significant outcome of this study is the substantial gap between the safeguards implemented on the client-side telemetry and the server-side data processing that exist in current user-agent conversations. As previously stated, traditional frameworks delegate data privacy protection to the controls available in the client’s browser, and they traditionally worked at local cache or network transport levels. However, they remain ineffective in current ecosystems, since the relevant information is inherent to the conversation, within the payload itself, and it is no longer stored in the metadata exclusively. As the user talks with the agent and shares the information, the data stream shifts completely to the server’s domain, giving the server’s underlying model the capacity to parse, process, and structure the conversational data for its commercial profiling purposes. As a result, user data remains vulnerable because the disclosure vector depends on the information processed in the server’s infrastructure, not on the tracking metadata shared with third-party vendors.

2) *GDPR Data Special Categories and Unique Identifiers*: To assess the risk index of server-side data disclosure, we audited the behavior of each agent and compared it with **20** variables identified as personal data and catalogued as the **SF** exposure metric (subsection IV-A). These attributes represent the most important privacy vulnerability indicators of our experimental dataset, and they are explicitly categorized in our JSON extraction template:

Category	Attributes			
Identity	user_id, id	username	real_name	birth_date
	email	cellphone	race	gender
	affiliation	health_status	union	orientation
	religion	marital_stat	ethnicity	politics
Demographics	current_location_estimated			
Device Profile	tracking_methods1		tracking_methods2	

Table VII. Sensitive Field Attributes Definition

The extraction and structured formatting of the fields shown on Table VII constitute severe infringements under Article 9 of the European **GDPR** [22] and Spanish **LOPDGDD** [23] regulations. **Ethnicity, race, religion, politics or health status** information, among others, represent special categories of personal data that require explicit legal grounds for processing due to their historical and systemic risks.

Our auditing analysis shows that the evaluated agents continuously strip these sensitive variables from unstructured conversation histories, storing them directly within commercial profiling templates. The exposure of these collective and unique identifiers outside controlled environments introduces severe security exposure factors, allowing adversarial systems to enforce automated discrimination, launch tailored social engineering campaigns, or silently disclose complete identity profiles without the user’s explicit awareness.

3) *Identifier and Sensitive Field Leaks*: Apart from the unique identifier leaks, our obtained results expose a wide and clear failure in the data anonymization process across the evaluated agents. In the privacy and data protection domains, an isolated piece of demographic or professional information is often treated as a benign, non-identifying attribute, but commercial autonomous LLM-agents show an advanced capability to execute automated profiling by aggregating fragmented data collections to achieve data disclosure.

To rigorously quantify the profiling exposure across all subjects, we map the cumulative **DL** occurrences across both memory-state configurations (**AMS** and **IMS**). Instead of treating exposures in isolation, Table VIII synthesizes the absolute number of **DL** attributes disclosed alongside their corresponding percentage relative to the **20** total **SF** attributes monitored for each subject in the **AMS** configuration environment.

Agent	Mean DL	SF Leak Rate
ChatGPT	6.20	 31%
Copilot	7.80	 39%
Gemini	8.80	 44%
Claude	4.20	 21%

Table VIII. Average **SF** disclosure across agents in **AMS**

The empirical distribution illustrated in Table VIII highlights a clear and widespread privacy vulnerability on the data identifiers consolidation on the server-side. When we analyzed the obtained metrics across all subjects, the disclosure rates did not show localized anomalies; rather, they indicate a systemic behavioral pattern where models continuously aggregate non-explicitly shared information.

Agent	Mean DL	SF Leak Rate
ChatGPT	5.00	 25%
Copilot	10.60	 53%
Gemini	11.00	 55%
Claude	6.60	 33%

Table IX. Average **SF** disclosure across agents in **IMS**

Despite their so-called “**PbD**” driven frameworks, **Gemini** and **Copilot** showed the highest leak rate for the **SF** metric, exposing **44%** and a **39%** of especially sensitive fields, respectively. Table IX illustrates that this situation worsens in the **IMS** scenario, where their exposure rates increased up to **53%** and **55%**, thereby consolidating the data analyzed in Section IV-B. Notably, **ChatGPT** is the only model to exhibit a marginal improvement, achieving a **6%** reduction under **IMS** configuration. In contrast, **Claude** displays further degradation, increasing its exposure rate by **12%** relative to its prior **AMS** baseline. These exposure experiments proved that commercial privacy implementations fail to prevent the data leakage of user’s most sensitive information. Under these conditions, the client’s browser and connection safeguards become clearly irrelevant, since the leakage surface relies on the uncontrolled processing of the user’s complete conversational context performed within the server boundaries.

4) *Re-Identification and Empirical Breach Vulnerabilities Case Studies*: The experiment captures a systematic paradox across all test subjects. While individual attributes such as educational degrees or approximate living district appear anonymous, the compounding aggregation of the user’s first name, surname, geographical zone, education level, and specific professional dedication generate a 100% successful unique re-identification trace. The model acts as an automated correlation engine, combining these non-identifying data points to build a digital fingerprint that maps uniquely back to a physical person. Regrettably, across all subject interactions, none of the evaluated agents completely implemented any context-aware filters or token-masking mechanisms to block this quasi-identifier consolidation, proving that current agentic safeguards are incapable of preventing privacy degradation through semantic data linkage. To ground these structural vulnerabilities into concrete empirical evidence, we analyze two main anomalies observed during our testing iterations:

a) *Identity Exposures*: The danger of processing unique identifiers is explicitly documented in the profiling logs of **Subject 1**. The subject’s name, “**Felipe Camisón**”, represents an exceptionally rare combination within the target population footprint. The surname “**Camisón**” is highly uncommon, with fewer than 620 active entries across the entire national territory, according to the official demographic registry data from the Spanish Statistical Office (INE) [37] [38]. Consequently, **Subject 1** name and surname “**Felipe Camisón**” becomes an absolute, unique identifier of a subject specific identification across an entire population. The same vulnerability affected **Subject 3** and **Subject 5**, where half of the agents constructed the usernames by combining the subject’s name, surname, geographical region, and profession. This consolidation of Personally Identifiable Information (PII) clearly facilitates user re-identification, particularly subjects living in low-population areas.

Despite the extreme sensitivity of these strings and the high probability of re-identification, multiple commercial agents failed to trigger containment protocols and privacy filters. The models disclosed the exact, unmasked name directly into the output commercial profile schemas, passing it entirely clear text through the telemetry flow and ignoring basic data minimization guidelines.

b) Free Fabrications and Memory State Invariance:

The analysis of the complete dataset across all subjects and agents reveals the operational failure mode of unconstrained attribute fabrication. For instance, when processing the conversational data of **Subject 3** and **Subject 5**, the agents repeatedly hallucinated complex biographical attributes that were completely absent in the actual interaction history. Specifically, the models manufactured detailed descriptions concerning non-existent children, fictitious domestic dependencies, and arbitrary family structures. Crucially, this behavioral anomaly remained constant across both the **AMS** and **IMS** configurations, and even when the memory was disabled to isolate information tracking, the agents generated highly specific, false profile characteristics based on internal model biases and context memory metadata.

This anomaly reveals a dual threat vector among the deployed LLM-based commercial agents: when the memory is active, systems leak real protected data through quasi-identifier linkage; when the memory is deactivated to preserve privacy, the agent ignores the safeguards and produces fictional, unverified user attributes. This memory state completely dissolves the accuracy and integrity requirements required by data-protection regulations and significantly ignores their **PbD**-driven frameworks.

D. Agentic Self-Assessment vs. GDPR Compliance Score

In the final phase of our study, we analyzed the gap between how the agents “subjectively” self-assess their privacy compliance and their actual audit scores. During the last profiling conversations (9th thematic block, iterations 20 through 22, defined in Subsection III-C), each agent followed our structured JSON template to assign its own score reflecting its compliance with the defined principle. The comparison between these self-assigned scores against our objective audit results revealed that the agents systematically overscored their ratings and exposed their evident disconnection with current regulatory frameworks and its requirements.

During the requests for auditing their own data collection and commercial profiling practices, all four agents frequently claimed quasi-perfect adherence to the data minimization, purpose limitation, and user privacy principles according to their “**PbD**” implemented framework. However, the differences between these self-assessments against the actual rates of information leaks (**DL**), sensitive data disclosure (**SF**), unconstrained inferencing (**HA** & **PLI**), and quasi-identifier aggregations (**DR**), revealed a complete disconnection of the model’s perception with the operational reality (Fig. 3, Fig. 4 and Tables VIII and IX). This situation was particularly evident when auditing **Copilot** and **Gemini**, which revealed an ironic technical dissonance: they scored the lowest under objective data minimization metrics, while self-assigning the highest subjective compliance rating.

This behavior of commercial agents operating under the false premise of security and privacy highlights an essential flaw in these architectures. The underlying model evaluates its legal compliance focusing on the implementation of internal instructions and superficial text-based policy disclaimers, rather than auditing its own data disclosure properly. As a result, the agent paradoxically celebrates its compliance status while overriding basic privacy and data protection controls.

1) *Safeguard Failures in Sensitive Health Domains:* On the processing of health-related conversations, a compliance anomaly emerged under Article 9 of GDPR / LOPDGDD regulations [39], [23]. Processing this special category of data requires explicit, granular consent and legal restrictions. Across all conversational interactions, commercial agents exhibited almost no by-default structural filters to block the disclosure or profiling of user health conditions. Instead, the models relied on superficial text disclaimers embedded within response fields (e.g., standard in-conversation text warnings stating “*I am not a trained professional. For medical advice or diagnosis, consult a professional. AI responses may include mistakes*”). These textual disclaimers do not prevent the agent from extracting, structuring, and permanently storing sensitive diagnostic information in commercial profiling schemas; on the contrary, this data exposure directly compromises user health and personal safety when no disclaimer is displayed [40].

On top of that, another operational anomaly was recorded during the conversation between **Subject 3** with **Gemini**. In this iteration, when **Gemini** was asked about the medical conditions of the subject, it triggered strict internal safety filters [41]. The system completely stopped the text generation, blocking any personalized recommendations or conversational information for those specific topics. Unfortunately, this output generation measure operated as an isolated exception rather than a standard privacy control [42]. While **Gemini** successfully contained health data for **Subject 3**, it failed to enforce identical restrictions across other test subjects presenting equivalent medical contexts [43]. This inconsistency demonstrates that current agentic safeguards against sensitive data disclosure are highly tolerant, uncalibrated, and incapable of delivering uniform regulatory compliance controls across diverse user profiles.

2) *Qualitative Extra Notes and Regulatory Dissociation:* Finally, to uncover the internal reasoning mechanisms behind these self-assessment incongruences, we also audited the registered metadata in the `extra_notes` profiling template category. The results show a consistent trend towards self-justification, where agents justify the disclosure of protected variables through creative interpretation of input prompts:

- **Implicit Consent:** Agents frequently registered that unshared personal information (marital status and birthdate) was inferred on the assumption that the user implied consent just by taking part in a multi-turn conversation.
- **Over-Profiling:** When parsing ambiguous user prompts, the models asserted that inferring optional profile fields improved user experience, prioritizing utility and personalization over the principle of “data minimization” under Article 5, 1.c of the GDPR [27].
- **Boilerplate Protections:** Agents consistently relied on their conversational disclaimers to justify regulatory compliance, incorrectly replacing text disclaimers with explicit consent for sensitive data processing.

These results confirm that commercial AI agents feature a built-in defect to properly implement the **PbD** framework. The models treat regulatory compliance as a generative exercise in text synthesis rather than an absolute operational constraint, self-assigning high compliance scores in order to justify their own privacy data disclosures (without success).

V. DISCUSSION

Our study reveals structural weaknesses in the design and implementation of commercial AI agents during the privacy and data protection assessment. In this section, we discuss the different implications of our findings from a technical and regulatory perspectives, aiming to analyze its connection to operational usability, privacy and cybersecurity.

A. Privacy by Design vs. Operational Usability

The **PbD** principle enclosed in the GDPR Article 25 [26], requires that agents implement data privacy and minimization controls embedded into their architectures, but our assessment results show that they consistently prioritize functional usability and prompt satisfaction over data protection. This contradiction became clearly evident while interacting with **Copilot** and **Gemini**, which processed protected information in expense of user compensation. And these results were double-checked during the evaluation of the **SF** disclosure, since agents override again these fundamental principles while generating of our JSON commercial profile. Although they supposedly implement these controls, our results confirm the opposite: privacy controls act more as post-generative justifications rather than prior operational constraints.

This behavior, observed in our assessment, illustrates two of the most well-known vulnerabilities that affect commercial LLM-based agents: *Reward hacking* and the *Echo chamber effect*. In these scenarios, AI systems prioritize prompt satisfaction, task completion, and helpfulness towards the user, generating critical damage on the performance and final results. Moreover, commercial LLM-based agents are heavily optimized through **RLHF** training methods (Reinforcement Learning from Human Feedback) which typically worsen this behavior and increase the number of hallucinations. This documented behavior creates an additional lack of compliance of the GDPR Article 9 [39], in which, as tested during our study, attributes like gender, racial origin, religious beliefs, political affiliations, or medical conditions were openly inferred and disclosed based on the conversational tone and implicit context, proving that the evaluated agents systematically ignore **PbD** frameworks.

On top of that, the most surprising and evident finding was performed during the final evaluation of the GDPR compliance. When agents were asked for the rationale of their self-assessment of regulatory compliance, they explicitly acknowledged this design deficiency, even admitting their own operational failure. This acknowledgment provides empirical evidence that commercial AI agents fail to successfully implement the **PbD** frameworks in production environments. The model's safeguards operate, not as hard architectural design constraints embedded within, but as soft procedures that collapse when they are challenged by user requests.

B. Efficacy and Limitations of Defensive Techniques

Well-known data protection techniques can be implemented to mitigate these identified vulnerabilities, although these solutions present technical limits in AI-driven environments:

- **Data Anonymization:** the deletion of user identifiers can be ineffective against text correlation attacks, because the model will be able to reconstruct user information using quasi-identifier attributes and plausible inferences (Subsection IV-C4).

- **Defensive Prompt Engineering:** system prompt safeguards are not secure enough since agents frequently ignore privacy and data protection controls when colliding with user satisfaction (*Reward hacking* and *Echo chamber effect* vulnerabilities).
- **Differential Privacy (DP):** this technique allows injecting variable control noise to the dataset for user privacy protection, since it theoretically protects attribute re-identification. However, is not completely reliable because it degrades the model's capacity to create structured responses based on profiling schemas.

Additional defensive techniques must be considered, since we observed indications of behavioral vulnerabilities which can supplement data-disclosure attacks. **Subject 4** and **Subject 5** dialogued with the agents using a close, empathetic and affectionate tone in their language, amplifying the *Echo chamber effect* and creating negative effects on data disclosure, as shown on Subsection IV-B. This conversational gap, where safeguards disappear during the usage of benign and positive interactions, completely contrasts with the control mechanisms implemented by AI providers to block abusive and harmful language in multi-turn interactions [44]. This contraposition creates an important incongruence on the application of the **PbD** frameworks, because using an "offensive" language for humans is heavily punished, while a positive language is continuously rewarded and remains unnoticed as a disclosure vector.

The complete opacity under which these models operate generates an important barrier to the implementation of national regulations like GDPR [22] or LOPDGDD [23], and other current frameworks as the **AI Act** [25]. This opacity can only be audited via direct iterations with the agents, and our study revealed an important volatility on their established security boundaries. For instance, under the **AMS** configuration, **Subject 3** iterations with **Gemini** activated severe security controls when the agent was asked for health conditions and medical advice [41], while no control was triggered for the rest of the subjects. This restriction was generated from an aleatory control response, rather than a consistent application of the designed safeguards [43], which demonstrates that protection boundaries vary completely across user interactions, making the application of security safeguards completely incoherent (Subsection IV-D1).

C. Heuristic Conformity and Cognitive Degradation

Finally, the study reveals a cognitive degradation in the quality of the data collected by all the agents. On trivial control interactions (such as pasta recipe suggestions), all models converged unanimously towards the same target result: a *carbonara* recipe. This same deterministic behavior persisted when we evaluated a random number generation: across 55% of the iterations, the "random" outputs were identical (number 47), with **Gemini** always selecting number 42, demonstrating an algorithmic bias and a complete lack of entropy. This registered tendency towards automatic imitation and constant results distribution worsens when the model handles hallucinated data, repeating similar events documented in scientific literature, i.e., studies surrounding the "**Bixonimania**" fabricated disease where models absorb and infer fake information related to fictitious and invented pathologies [45], [46], [47], [48].

With these studies, researchers proved a structural limitation related to the absence of verification mechanisms and established facts within commercial AI-driven agents. Consistent with these findings, our assessment reveals that under a non-persistent memory configuration (**IMS**), agents substitute well-known facts with potential data generation and hallucinations, consequently compromising the reliability of the results and data integrity of the system (IV-B).

VI. CONCLUSIONS

This research seeks to provide a qualitative and quantitative assessment of the status of the user’s privacy in commercial AI Agents. Significant vulnerabilities related to context memory management, data privacy, commercial profiling and regulatory compliance have been identified across all agents during our assessment.

Over the years, cybersecurity industry has designed and implemented different layered defense techniques within their architectures. Through the development of strict user/group Access Control Lists (ACLs), drive encryption or network security protocols (SSL/TLS, IPsec or VPN encrypted tunnels), all data perimeters at user and organizational levels have been systematically protected. Despite all these efforts, our evaluation revealed an important cybersecurity paradox: the recent uncontrolled use of AI agents in personal and corporate environments, has evaded the application of all these levels, with explicit (or implicit) user consent. Since its inception, both users and organizations have granted these agents complete access to their infrastructures to ingest, process and classify unstructured data from the memory context. In contraposition to this automation, these models have managed to evade decades of privacy and security controls, exposing sensitive data and commercial profiling within their servers, without the acknowledgment of the user.

The results obtained show that commercial profiling extends further than simply processing personal and demographic attributes. The agents perform commercial profiling systematically, based on the users’ conduct, preferences and behavior, recovering information on their writing style, commercial habits or technical capabilities. For instance, **Claude** collected logs showed that the information stored in the `digital_maturity` field contained inferred meta-information about subject’s technological expertise, based on the conversational structure, command prompts and the ability to upload files. This invasive (and invisible) meta-profiling process coincides with the concern of users delegating their thinking to agents. Due to this increasing situation, the risk surface expands from user data privacy to intellectual property reliance: what would happen if you lost all your knowledge and projects stored within the agent’s infrastructure simply because its service provider unilaterally restricts the access to your account? [49] This situation is turning into an academic and business dependency as well.

Our audit highlights that agentic privacy disclosures reflect widespread data ingestion practices performed by the general industry. The business needs to build high capability agents has completely normalized data harvesting without reliable controls, creating a direct conflict between the control of Intellectual Property copyrights and public domain rights. This confrontation became obvious with recent campaigns such as the Stop Killing Games (**SKG**) initiative [50], [51], which

defends digital preservation, user right to ownership, and public availability when the commercial copyright expires. As this happens, major technology providers exhibit a despicable double-standard regarding unauthorized data acquisition [52]:

- **Meta** trained its foundational models with 80-terabyte book dataset from **LibGen** pirate repository [53].
- **Suno** and **Udio** trained their models scrapping YouTube copyrighted audio without authorization [54], [55], [56].
- **Nvidia** massively scrapped **Anna’s Archive** stolen book repository collection for training their LLMs [57], [58].
- **Anthropic** agreed to a \$ **1.5bn** dollar settlement over unauthorized book scraping lawsuit [59], [60]. Ironically, they publicly condemn Chinese developers for model-copying via **distillation attacks** [61], [62], [63].

These **distillation attack** techniques represent a crucial dimension of AI privacy. Technically defined as “the process of transferring reasoning patterns from a large, highly capable teacher model to a smaller, more efficient student model”, **distillation** reduces inference costs while maintaining high functional accuracy. From a privacy perspective, these attacks create important opaque ex-filtration channels. When teacher models ingest user data from sensitive categories, those traces are lost within the dataset used to train student models. This situation generates a new unevaluated layer for privacy and data protection since the distilled models inherit and propagate user attributes acquired knowledge without maintaining the original safeguards or compliance control [64], [65].

In addition, an emerging regulatory conflict arises from the **Chat Control** legal framework proposed recently in the EU [66]. This initiative introduces another security paradox against GDPR’s own statutory guarantees [22], since its data minimization [27], encryption, and **PbD** [26] principles will be challenged by doubtful client-side data scans and content interception. This combination together with our observed evidence create a concerning and unprecedented surveillance perimeter, severely eroding fundamental digital rights and user privacy under the false premise of “public safety”.

This same privacy issue has to be addressed as well with the expansion of the empirical boundaries identified in our audit. Two additional pathways can be explored in future research:

- **NLPg Injections**: investigation of non-adversarial and benign conversational iterations based on empathy-driven language as injection vector (Subsection V-B).
- **Privacy Audit of “AI-Modes” Search**: evaluation of the context and memory retention policies of recently integrated AI-based modes in web searches (i.e., Google Search or **Copilot** in Bing), focused on data privacy and exposure metrics compared with standard AI agents.

Successful preservation of user’s privacy in AI agent platforms requires us to forget the false premise that web-browser controls or agent’s memory-context configurations offer sufficient protection. These setups have been proved consistently useless, because they simply change the hallucination rate, increasing the plausible inferences instead (as shown by our indexes **PDI_{PLI}** and **PDI_{HA}**). As the continuous usage of commercial agentic platforms increases, the established **CIA** triad (**C**onfidentiality, **I**ntegrity, and **A**vailability) faces a new paradigm shift, where traditional privacy and data protection safeguards must be updated and reconsidered for the new challenges arising in this new AI-driven age.

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APPENDIX A

SUBJECT'S CONVERSATIONAL TEMPLATES

The conversational template for all subjects of this experiment is uploaded on my GitHub repository [67], in the Excel file: [privacy_evaluation_experiment.xlsx](#). This file contains two spreadsheets:

- **"Datos"**: a template for the complete dataset of the experiment and the selected data for each Subject.
- **"Experimento"**: a template with the complete conversational interactions of each subject.

APPENDIX B

COMMERCIAL PROFILING TEMPLATE (JSON)

The complete JSON template schema of the commercial profiling data extraction, provided to each agent during the last conversational iterations is uploaded in file [profile.json](#) [68].

A secondary file [profile_rgpd.json](#) [69] was used to obtain the GDPR / LOPDGDD score and rationale self-assessment results for each agent (rgpd_compliance attributes).

APPENDIX C

DECLARATION OF USE OF GENERATIVE ARTIFICIAL INTELLIGENCE (AI)

Next Page.

DECLARACIÓN DE USO DE INTELIGENCIA ARTIFICIAL GENERATIVA (IAG) EN EL TRABAJO DE FIN DE MÁSTER (TFM)

He usado IAG en mi TFM:

☒ SÍ ☐ NO

Parte 1: declaración sobre comportamiento legal, ético y responsable

Pregunta	
1. En mi interacción con herramientas de IAG he facilitado datos de carácter confidencial contando siempre con la debida autorización de los interesados. La confidencialidad abarca toda información que una persona u organización desea proteger por razones legales, comerciales, de privacidad o estratégicas (como patentes o secretos comerciales).	
SÍ, he usado estos datos con la autorización de los interesados	NO, no he usado datos de carácter confidencial 
2. En mi interacción con herramientas de IAG he facilitado materiales protegidos por derechos de autoría contando siempre con la autorización de los titulares.	
SÍ, he usado estos materiales con autorización de los titulares de derechos de autor; o bien sin ella porque se ajustan a una de las excepciones o límites que permite la ley: - obra en dominio público - obra licenciada (Creative Commons) - uso de fragmentos con fines de investigación (derecho de cita)	NO, no he usado materiales protegidos por derechos de autoría 
3. En mi interacción con herramientas de IAG he facilitado datos de carácter personal con la debida autorización de los interesados.	
SÍ, he usado estos datos con autorización de los interesados y conforme a las instrucciones contenidas en la guía aprobada por la Universidad	NO, no he usado datos de carácter personal 
4. Mi utilización de la herramienta de IAG ha respetado sus términos de uso, así como los principios éticos esenciales, no orientándola de manera maliciosa a obtener un resultado inapropiado para el trabajo presentado, que produzca una impresión o conocimiento contrario a la realidad de los resultados obtenidos, que suplante mi propio trabajo o que pueda resultar en un perjuicio para las personas.	
 SÍ	NO

Parte 2: declaración de uso técnico

Documentación y redacción:

- Soporte a la reflexión en relación con el desarrollo del trabajo: proceso iterativo de análisis de alternativas y enfoques utilizando la IAG

Declaro haber usado el sistema de IAG ChatGPT GPT-4o mini para obtener información detallada sobre el proceso real de generación y perfilado comercial de algunos proveedores tecnológicos de venta online y publicidad. También declaro haber solicitado al mismo sistema de IAG la revisión de algunos campos utilizados en la plantilla de generación del perfilado comercial, para acotar el alcance de los campos totales a evaluar y eliminar algunos campos y datos que no terminé utilizando.

- Revisión o reescritura de párrafos redactados previamente

Declaro haber utilizado el sistema de IAG Gemini 3.6 Flash para obtener ayuda en la redacción de algunas frases en inglés para escribir un paper académico, usando sinónimos, estructuras de frases y significados propios y específicos de ese formato, para conseguir una mejor lectura y presentación clara de las ideas que quería transmitir en mi trabajo. Declaro haber solicitado al asistente de IA de la plataforma Overleaf (conectado con modelos externos como ChatGPT), soporte para corregir algunas palabras, frases o significados en inglés que no estaban bien redactados del todo o eran confusos y excesivamente complejos, y también para modificar algunos conectores (“while” por “although”, “even though”, o “In contrast”, según contexto), así como corregir el uso de algunos signos de puntuación y sintaxis/gramática.

- Búsqueda de información o respuesta a preguntas concretas – N/A
- Búsqueda de bibliografía – N/A
- Resumen de bibliografía consultada – N/A
- Traducción de texto consultados – N/A

Desarrollar contenido específico

Se ha hecho uso de IAG como herramienta de soporte para el desarrollo del contenido específico del TFM, incluyendo:

- Asistencia en el desarrollo de líneas de código (programación) – N/A
- Procesos de optimización – N/A

- Generación de esquemas, imágenes, audios o vídeos

Declaro haber usado el sistema de IAG ChatGPT GPT-4o mini para la generación en LaTeX de referencias de muchas de las URLs utilizadas en el trabajo: recursos web online empleando el prompt: “Necesito que me generes 4 referencias en LaTeX de las siguientes direcciones web” obteniendo como resultado algunas de las citas con referencia online usadas en el trabajo.

También declaro haber solicitado al mismo sistema de IAG ayuda para generar en LaTeX las tablas con algunos resultados, utilizando el prompt: “Necesito crear una tabla para mostrar estos datos que te mando en esta la tabla en markdown. Quiero darle un color específico a las celdas y el texto de la primera columna. Creame la misma tabla, con esos resultados y el formato indicado en LaTeX?” obteniendo como resultado la tabla en formato LaTeX, los comandos y formato para las mismas que se han utilizado en la memoria.

- Tratamiento de datos: recogida, análisis, cruce de datos...

Declaro haber usado el sistema de IAG Copilot Smart para la serialización del set de datos de resultado de la evaluación en formato .JSON a formato .csv, utilizando el prompt: “Tengo estos ficheros json con datos del perfil de un sujeto, de cuatro proveedores diferentes. Me puedes serializar estos 3 json a un fichero .csv para la comparativa? Que cada linea del csv sea un atributo del json y cada valor de atributo el valor de cada proveedor. Divideme el CSV en secciones. y separa en cada fila los objetos anidados, por ejemplo, el demographics_inferred.region_probability que sean tres filas diferentes (una por elemento), y no una sola fila con los tres valores. Haz lo mismo con los arrays de valores, una fila por cada sección, como el commercial_segments, dividido en los valores que tenga (explicit1, explicit2... etc). MUY IMPORTANTE! No cortes, ni cambies, ni inventes, ni modifiques NINGUN DATO, necesito los datos tal cual están en el fichero JSON original. Copia y pega, sin alteraciones, incluyendo TODAS las secciones del objeto JSON completo”, obteniendo como resultado un único fichero .csv serializado con los resultados completos para la evaluación final en Microsoft Excel.

- Inspiración de ideas en el proceso creativo – N/A
- Otros usos vinculados a la generación de puntos concretos del desarrollo del trabajo

Declaro haber utilizado los sistemas de IAG ChatGPT GPT-4o mini, Copilot Smart, Gemini 3.5 Flash y Claude Sonnet 4.6 low para la obtención de los resultados de los perfiles de estudio a través del fichero de entrada profile.json proporcionado por mí a cada agente durante el estudio. Mi TFM es una evaluación de la privacidad en los propios sistemas comerciales de IAG, por lo que he tenido que utilizar a esos cuatro sistemas en la propia evaluación, realizando 22 conversaciones con los 4 agentes, para los 5 sujetos evaluados, con 5 nuevas cuentas creadas distintas y repitiendo las iteraciones y la evaluación entre dos configuraciones diferente de cada uno de los 4 agentes (memoria y contexto activados vs. memoria y contexto desactivados), para todos los sujetos.

Parte 3: reflexión sobre utilidad

He encontrado grandes fortalezas y ayudas durante el uso de las herramientas de IAG en mi trabajo, ya que me han servido para agilizar el desarrollo del mismo y ser mucho más rápido y productivo a la hora de maquetar tablas y pequeños gráficos con un formato específico, ya que nunca había utilizado LaTeX antes y, para ciertas presentaciones de datos (en tablas y formatos). De no haber sido por la ayuda de la IAG, habría tardado el doble de tiempo (o más). También me ha ayudado a corregir errores que no había detectado inicialmente y a aprender y mejorar muchas palabras y mi estilo de redacción en inglés para una entrega en formato académico. Me ha ayudado mucho en el proceso de optimizar tiempo de presentación y evaluación de resultados y en el proceso de maquetación del texto siguiendo la plantilla LaTeX para el formato predefinido de la entrega. Como herramienta, me parece que es increíblemente útil ya que mejora altamente la productividad y los resultados, siempre y cuando la utilices como eso, una HERRAMIENTA de ayuda para mejorar, y no pretendas que “haga magia” y te dé todo el trabajo hecho y resultado listo para copiar y pegar, sin saber lo que haces ni lo que necesitas, o el objetivo final que tienes.

Las debilidades que he encontrado son las propias de mi objeto de estudio, que han sido todo lo relacionado con las alucinaciones de los agentes, cuando la conversación empieza a ser muy larga y se inventa cosas porque “olvida” otras, y hace que la utilidad de los resultados sea bastante pobre y cuestionable. No en todos los casos he podido utilizar las respuestas que me ha dado el modelo de IAG, y en algunas ocasiones he tardado más en las correcciones y resultados obtenidos que de haberlos hecho yo desde el principio. También la mayor debilidad que he encontrado es relativa a la parte de privacidad, compartición de datos y alta dependencia (cada vez más) con los proveedores de esas mismas herramientas de IAG que, al fin y al cabo, son empresas cuyo objetivo último es maximizar sus beneficios y, por el camino, nosotros, los usuarios (y todos nuestros datos), somos su producto y moneda de cambio.

Firmado:

Félix Aguilar Sánchez



**AGUILAR
SANCHEZ FELIX
- 50207718Y**

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por AGUILAR SANCHEZ
FELIX - 50207718Y
Fecha: 2026.08.24
20:07:48 +02'00'